SecureVoice: Closing the Studio-to-Real-World Domain Gap in Anti-Spoofing for Voice-Authenticated Transactions

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***Abstract*—Voice-authenticated transaction systems increasingly rely on speaker verification to answer "is this the enrolled user's voice?" but rarely ask the different question, "is this a real human?" We build a two-layer authorization pipeline—ECAPA-TDNN speaker verification plus AASIST-L anti-spoofing—into a voice-ordering system and measure, rather than assume, its security. Across 800 trials over 20 enrolled speakers on ASVspoof2019 LA, 54.0% of synthetic voice-conversion/TTS attacks clear the speaker-verification gate, while the anti-spoofing layer independently flags 87.0% of them with zero false alarms on genuine or impostor speech, cutting approved attacks to 7.5% at typical transaction tiers. We then reproduce the anti-spoofing model's published benchmark (0.99% EER, matching the originally reported figure) and document a severe practical failure: on our own real phone recordings, the same model with the same frozen threshold misclassifies 100% (8/8) of genuine speech as synthetic. We diagnose this as a studio-versus-real-world domain gap—ASVspoof2019's bona fide class is anechoic-chamber audio with a 16× lower noise floor and 10× more high-frequency energy than a phone recording in an ordinary room—rule out codec, resampling, and gain artifacts as the cause, and show that direct domain adaptation on a small set of real recordings reduces the real-world false-positive rate from 8/8 to 2/8 at a measured, bounded cost to benchmark EER (2.00% → 8.12%). We additionally quantify how the same model degrades under VoIP-style channel loss (0.99% → 14.00% EER) and under short streaming windows (0.99% → 36.43% at 250 ms), and report all findings—including negative results—under a fixed, development-set-only threshold-selection protocol.**

***Index Terms*—voice authentication, anti-spoofing, ASVspoof, speaker verification, domain adaptation, AASIST, ECAPA-TDNN, deepfake speech detection**

I. Introduction

Personal voice assistants and voice-authenticated transaction systems increasingly gate real financial actions—placing orders, moving money, unlocking doors—behind a voice check [1]. Smart-speaker adoption has grown rapidly and voice is now marketed as a convenience layer atop e-commerce and banking flows, which means a successful attack on the voice channel is no longer merely a privacy nuisance but a direct financial-fraud vector. The dominant technical approach to securing that channel is speaker verification (SV): compare an incoming utterance against an enrolled voiceprint and approve if the cosine similarity of speaker embeddings clears a threshold. SV answers "is this the right voice?" It does not answer "is this a real human speaking, or an AI-generated clone?"—a distinct question that a determined attacker can now target cheaply with modern text-to-speech (TTS) and voice-conversion (VC) tools, several of which are freely available and require only a few seconds of reference audio.

This gap is not hypothetical. A clone is explicitly optimized to sound like the target speaker, so a successful clone should, by construction, score well on SV—its similarity should rise, not fall, relative to an unrelated impostor. Prior surveys of the personal-voice-assistant (PVA) threat landscape [1] identify voice authentication spoofing as a mature but still-unresolved research area, and note that commercial assistants rarely combine spoof detection with access control in practice. Meanwhile the anti-spoofing research community has produced increasingly accurate countermeasures—sub-1% Equal Error Rate (EER) is now routine on the standard ASVspoof benchmark [5]—creating an appealing but, as we show, premature assumption that bolting such a model onto an SV pipeline solves the problem.

We built *SecureVoice*, a voice-ordering system secured by (i) ECAPA-TDNN speaker verification [6] and (ii) an added anti-spoofing layer using AASIST-L [5], a compact (85k-parameter) graph-attention network benchmarked on the ASVspoof2019 Logical Access (LA) corpus [2]. Rather than assume the resulting system is secure because each component is individually well-benchmarked, we measured it end to end—against synthetic attacks, against degraded audio, against short streaming windows, and, critically, against our own real voices recorded on an ordinary phone in an ordinary room. That last test, originally intended only as a pre-demonstration sanity check, produced this paper's central finding: a model reporting 0.99% error on its own benchmark misclassified every one of our real recordings as synthetic, with high confidence, using the same frozen operating threshold used everywhere else in this work.

This paper's contribution is therefore as much methodolog-

ical as architectural: we show the value of testing a voice-security pipeline against the conditions it will actually be deployed into, rather than stopping once each component clears its own published benchmark. Concretely, we contribute:

- a measured, controlled demonstration that SV alone is insufficient against synthetic speech, using an 800-trial genuine/spoof/impostor protocol with authorization outcomes computed under a real transaction tier policy;
- reproduction of a state-of-the-art anti-spoofing benchmark to validate our pipeline before trusting any downstream claim built on it;
- an independent reproduction and diagnosis of the studio-to-real-world domain gap on consumer-device audio, with control experiments ruling out codec, resampling, and gain artifacts as the cause;
- a lightweight domain-adaptation mitigation, reported honestly alongside a first attempt that failed, with its benchmark-accuracy cost quantified; and
- quantification of the same model's degradation under realistic VoIP channel loss and short streaming windows, relevant to continuous rather than one-shot voice authentication.

Section II situates this work against prior voice-spoofing and PVA security literature; Section III details the system and experimental protocols; Section IV reports and discusses results; Section V concludes.

II. Literature Survey

Voice authentication and its spoofing vulnerabilities have a long research history predating current PVA deployments [1], [20]. Five attack families are established: replay, impersonation, speech synthesis, voice conversion, and adversarial perturbation. Replay attacks play back a prerecorded genuine sample; synthesis and voice-conversion attacks instead generate new audio designed to match a target voiceprint, and are the family this paper concerns itself with, since modern TTS/VC pipelines make them cheap to mount at scale. The ASVspoof challenge series [2]–[4] standardized datasets, protocols, and the EER and t-DCF [19] metrics, distinguishing Logical Access (LA: synthesis/conversion attacks fed directly to the verifier) from Physical Access (PA: replay through a real microphone/loudspeaker channel). We scope this work to LA attacks; replay and liveness detection is a related but distinct problem with its own countermeasure literature [13], [14].

A. Benchmark Performance of Current Countermeasures

AASIST [5] is a graph-attention network operating directly on raw waveforms via a SincNet-style front end [18], building on the earlier RawNet2 [17] and RawGAT-ST [8] systems. Table I reproduces its reported comparison on ASVspoof2019 LA. Sub-1% EER is attainable, and notably the 85k-parameter AASIST-L variant outperforms systems two orders of magnitude larger, which is why we selected it for an on-device authorization setting. ECAPA-TDNN [6] is a widely adopted speaker-embedding architecture producing fixed-dimension embeddings compared via cosine similarity; we use it as SecureVoice's verification layer, matching common practice of pairing a verification embedding model with a separately trained countermeasure.

TABLE I
PUBLISHED SINGLE-SYSTEM RESULTS ON ASVSPOOF2019 LA EVALUATION, AS REPORTED IN [5]. REPRODUCED FOR CONTEXT; NOT OUR MEASUREMENTS.

System	# Param	min t-DCF	EER (%)
AASIST [5]	297k	0.0275	0.83
AASIST-L [5]	85k	0.0309	**0.99**
RawGAT-ST [8]	437k	0.0333	1.19
SENet	1,100k	0.0368	1.14
Res-TSSDNet	350k	0.0481	1.64
Raw PC-DARTS [9]	24,480k	0.0517	1.77
LCNN-LSTM-sum [10]	276k	0.0524	1.92

TABLE II
BENCHMARK-TO-REAL-WORLD EER DEGRADATION REPORTED BY [7]. THE FINAL ROW SHOWS THEIR FINDING THAT FIXED-LENGTH 4 S INPUTS UNDERPERFORM FULL-LENGTH INPUTS—DIRECTLY RELEVANT TO OUR STREAMING RESULTS (SECTION IV-F).

System	ASVspoof19	In-the-Wild	Degrad.
RawGAT-ST	1.23%	37.15%	30×
RawNet2	3.15%	37.82%	12×
12-model average	9.85%	60.10%	6×
Fixed 4 s input (avg)	18.89%	67.25%	3.6×

B. The Generalization Problem

A growing body of work questions whether these numbers survive outside the challenge's own studio-recorded data. Müller et al. [7] re-implement twelve architectures under a common protocol and evaluate them on an "In-the-Wild" set of 37.9 hours of found audio from 58 celebrities and politicians. They report EER degradations of 200–1000%, with several models collapsing to near chance (Table II). Critically for our work, they attribute this to the composition of the benchmark itself, observing that ASVspoof is built on the VCTK corpus [12], which "exclusively features professional speakers and has been recorded in a studio environment, using a semi-anechoic chamber." Related work by the same group [11] further suggests ASVspoof-trained models partly key on non-speech cues such as leading and trailing silence rather than the spoofing artefacts they are assumed to detect.

Our work is positioned deliberately with respect to [7]: we do not claim to be first to observe that anti-spoofing generalizes poorly. Rather, we contribute (i) an independent reproduction of the phenomenon on a different class of real audio—self-collected, consent-based consumer-device recordings representing a specific deployment context, rather than found celebrity media; (ii) measurement of the downstream security consequence inside a working authorization pipeline, which [7] does not address; and (iii) a measured mitigation with its accuracy cost quantified. Prior work establishes that the gap exists; we ask what it does to a deployed system and what closing it costs.

Complementary defense families target different threat models. Channel-artifact and second-factor defenses—detecting the sub-bass overexcitation characteristic of loudspeaker playback [13] or phoneme-timing liveness cues [14]—address replay rather than synthesis, and are complementary to, not substitutes for, the LA-focused defense studied here. Zhang et al. [15] demonstrate a practical black-box mimicry attack against a smart-home SV system, underscoring that SV alone is an actively targeted attack surface. Feng et al. [16] motivate continuous rather than one-shot voice authentication, a goal our streaming experiments (Section IV-F) show is nontrivial once a fixed-input model is forced onto short windows.

III. Proposed System and Methodology

A. Two-Layer Architecture

SecureVoice authorizes a voice-issued transaction through two independent checks that must both pass (Fig. 1). **Layer 2** (speaker verification) extracts a 192-dimension ECAPA-TDNN embedding and computes cosine similarity against a centroid of enrollment embeddings (three short utterances per user); similarity below 0.50 denies outright regardless of amount, and above that floor a trust score ($100\times$similarity) is compared against a minimum required for the requested transaction's risk tier. **Layer 2.5** (anti-spoofing, our addition) scores the same raw audio independently with AASIST-L; a synthetic-speech probability above a frozen threshold denies regardless of Layer 2's outcome, so a perfect voice match cannot by itself override a positive spoof detection. Both layers operate on 16 kHz mono input and add well under a second of combined latency on commodity hardware.

B. Threshold Discipline

Every detector threshold is chosen on a held-out development split and frozen before any evaluation-split or real-world number is produced, and is never re-tuned per condition afterward, mirroring the discipline the ASVspoof challenges impose on participants [3]. AASIST-L's threshold is selected once by sweeping the ASVspoof2019 LA development split for the EER operating point, then reused unchanged for every subsequent experiment—including the real-world recordings, the degraded-channel evaluation, and the streaming evaluation. This is what allows the failure in Section IV-C to be trusted as a genuine property of the model rather than an artifact of a threshold quietly re-tuned after the fact.

C. Vulnerability Measurement Protocol

ASVspoof2019 LA labels every spoof utterance with the identity of the speaker it targets, enabling a like-for-like enrolled-speaker comparison rather than a pooled score. For 20 randomly drawn speakers, each enrolled from three held-out bona fide utterances, we constructed three trial types: *genuine* (ten further held-out bona fide utterances of the same speaker), *spoof* (twenty TTS/VC attacks targeting that speaker), and *impostor* (ten bona fide utterances of a different speaker)—800 trials total. Each trial is scored by both layers using their independently frozen thresholds; we report the resulting similarity, spoof probability, and the authorization outcome both with Layer 2.5 disabled and enabled, so the marginal value of anti-spoofing is directly visible rather than inferred.

D. Real-World Domain Gap: Discovery and Diagnosis

After validating AASIST-L against its published benchmark (Section IV-B), we tested it on eight short assistant-style command phrases recorded on an ordinary smartphone in a normal room—not through the ASVspoof pipeline at all—purely as a sanity check before demonstrating the live system. To diagnose the resulting failure rather than merely report it, we compared long-term spectral energy distribution across five frequency bands and a noise-floor estimate (10th-percentile short-time RMS over 25 ms frames) across ASVspoof bona fide audio, ASVspoof spoof audio, and our recordings. We separately ran four control experiments—AAC compression at two bitrates, $16\leftrightarrow48$ kHz resampling, gain scaling, and mild additive noise (30 dB SNR) applied to clean ASVspoof bona fide audio—to test whether any single easily-corrected signal-processing artifact could reproduce the failure.

E. Domain Adaptation

We attempted two mitigations, both starting from the original pretrained checkpoint. **Fix 1** fine-tuned AASIST-L for two epochs on an ASVspoof2019 LA train subset (300 bona fide, 300 spoof) with synthetic room-reverb, additive-noise, and microphone-frequency-response augmentation applied with equal probability to both classes, so the augmentation distribution itself could not become a shortcut label. **Fix 2** instead mixed six of our eight real recordings directly into the same training mixture as additional bona fide examples, oversampled fifteen times each with light gain and noise jitter to avoid duplicate-sample memorization, while holding the remaining two clips out entirely so any improvement on them could only come from genuine within-domain generalization. Both were validated after every epoch on a fixed development subset to confirm the model had not collapsed into a degenerate predictor, and evaluated on a previously unseen 800-utterance evaluation subset for a comparable benchmark-cost figure.

IV. Results and Discussion

A. Speaker Verification Alone Is Insufficient

Table III summarizes the 800-trial protocol. ECAPA-TDNN is essentially perfect against a different human speaker (0.00% EER; no impostor clears the 0.50 gate) but far weaker against synthetic speech targeting the enrolled speaker (18.62% EER; 54.0% of attacks clear the same gate). The same embedding model, at the same threshold, behaves almost perfectly against one threat model and only moderately better than chance against the other, because it was never designed to distinguish "a different real voice" from "a synthesized copy of the target voice." In one representative case a clone scored 0.799 similarity against the enrolled user's own 0.693—the fake measuring as more authentic to the verifier than the genuine sample it was cloned from.

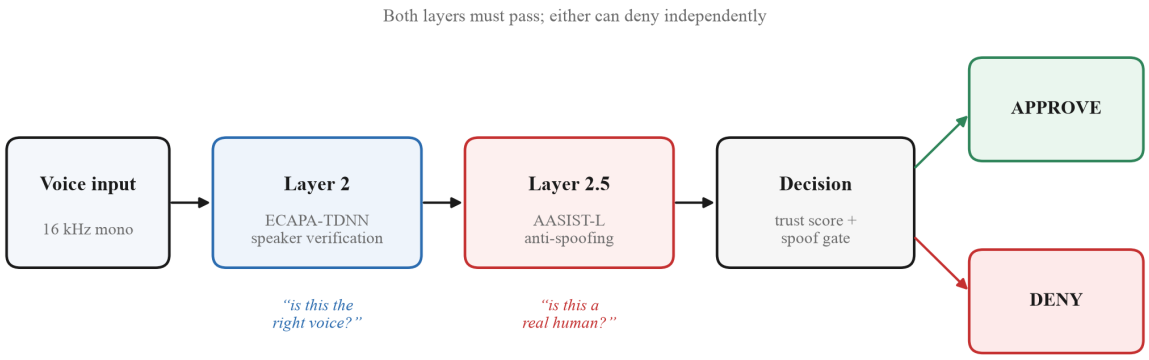


Fig. 1. SecureVoice two-layer authorization pipeline. Layer 2 asks whether the voice matches the enrolled user; Layer 2.5 (our addition) asks whether it is human at all. Either layer can deny independently, so a perfect voice match cannot override a positive spoof detection.

TABLE III
SPEAKER-VERIFICATION BEHAVIOR BY TRIAL TYPE ($n = 800$, 20 SPEAKERS).

Trial type	Mean similarity	Clears 0.50 gate
Genuine ($n = 200$)	0.726	99.5%
Synthetic clone ($n = 400$)	0.476	**54.0%**
Impostor ($n = 200$)	0.124	0.0%

TABLE IV
SYNTHETIC ATTACKS APPROVED, BY TRANSACTION TIER.

Tier	SV only	SV + anti-spoofing
<Rs. 500 / Rs. 500–2,000	54.0%	**7.5%**
Rs. 2,000–10,000	11.0%	**1.8%**
>Rs. 10,000	0.0%	0.0%

Adding the anti-spoofing layer flags 87.0% of synthetic attacks with zero false positives on genuine or impostor speech, cutting attacks approved at low-value tiers from 54.0% to 7.5% (Table IV, Fig. 2).

B. Benchmark Reproduction

Before trusting any downstream number we reproduced AASIST-L's published result on the full ASVspoof2019 LA evaluation set (71,237 utterances): **0.99% EER**, matching the originally reported figure [5] (Table I) against an 18.75% EER classical LFCC-GMM baseline fit on the same training split (LFCC-GMM development 9.38%/evaluation 18.75%; AASIST-L development 0.51%/evaluation 0.99%). We include the weak classical baseline not to contrast architectures for their own sake, but because its larger degradation under the same stressors usefully calibrates how much of AASIST-L's degradation is a property of the task versus the specific model. This agreement is what makes the failure in Section IV-C trustworthy as a property of the model rather than a symptom of a broken evaluation pipeline—a distinction that matters,

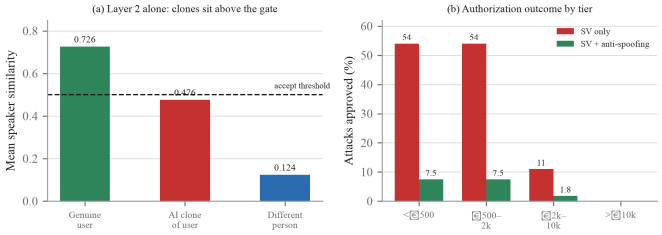


Fig. 2. (a) Mean speaker similarity by trial type: AI clones of the enrolled user sit above the 0.50 accept threshold, whereas a different human never does. (b) Share of synthetic attacks approved per transaction tier, with and without the anti-spoofing layer.

TABLE V
REAL-WORLD FALSE-POSITIVE RATE ON EIGHT BONA FIDE PHONE RECORDINGS, BEFORE AND AFTER DOMAIN ADAPTATION.

Sample	Original model	Domain-adapted
8 real recordings	8/8 flagged (100%)	**2/8 flagged (25%)**

since an unvalidated pipeline could equally produce a spurious 100% failure rate through a labeling bug.

C. A Benchmark-Perfect Model Fails on Real Audio

All eight real phone recordings were misclassified as synthetic, every one scoring above 0.98 probability against the frozen 0.9157 threshold (Table V). Spectral analysis (Table VI, Fig. 3) shows why: real phone audio carries a 16× higher noise floor and roughly 10× less 6–8 kHz energy than ASVspoof's anechoic-chamber bona fide class. The model had learned "studio-clean," not "human." Control experiments applying AAC compression, resampling, gain change, or mild noise to lab audio did not reproduce the failure, ruling those out as sole causes and implicating the recording environment and microphone chain as a whole. This is consistent with [7], obtained independently on a different class of real audio.

TABLE VI
SPECTRAL ENERGY SHARE AND NOISE FLOOR BY SOURCE.

Band	Lab bona fide	Lab spoof	Real phone
0–300 Hz	39.9%	46.8%	**53.8%**
6–8 kHz	0.38%	0.48%	**0.04%**
Noise floor	0.0043	0.0186	**0.0702**

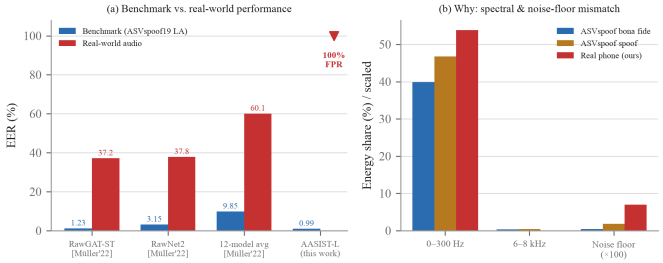


Fig. 3. (a) Benchmark versus real-world performance. Literature values from [7]; our AASIST-L benchmark EER is measured, and its real-world figure is a 100% false-positive rate on bona fide phone audio (no EER is defined without in-domain attacks). (b) Spectral and noise-floor mismatch underlying the failure.

D. Mitigation: One Failure, One Success

Fix 1 improved EER on synthetically-degraded validation audio (37.50% → 18.33% over two epochs) but did not transfer: real recordings remained 8/8 misclassified with scores essentially unchanged (0.93–0.99). We report this as a genuine negative result—approximated augmentation is not a substitute for real-domain data at this scale. **Fix 2** reduced real-world false positives to 2/8, and both held-out clips, never seen during training, flipped from >0.94 to <0.05 probability of being synthetic—a weak but genuine within-domain generalization signal. The cost was a bounded rise in benchmark EER on a held-out 800-utterance evaluation sample, from 2.00% to 8.12%. Importantly, the adapted model still detects real attacks (the spoof clip of Section IV scores 0.99999). This is a small-scale proof of concept—one phone, one room, one speaker—and not yet evidence of broad generalization.

E. Robustness Under Channel Degradation

Real voice traffic rarely reaches a back-end verifier as clean studio audio. We simulated an Opus codec round-trip at three bitrates crossed with Gilbert–Elliott burst packet loss at three rates on a class-balanced 3,000-utterance evaluation subsample, re-scoring both detectors with their already-frozen thresholds (Table VII, Fig. 4). AASIST-L's advantage over the classical baseline collapses from 15.6× on clean audio to 1.9× at the worst tested condition—the strong model still wins, but its practical margin nearly disappears exactly under the conditions a real VoIP transaction introduces.

F. Robustness Under Short Streaming Windows

Continuous authentication implies scoring short chunks rather than complete utterances, so we re-ran both detectors over non-overlapping 250, 500, and 1000 ms windows

TABLE VII
EER UNDER SIMULATED VOIP CHANNEL DEGRADATION.

Condition	LFCC-GMM	AASIST-L	Advantage
Clean	18.30%	1.17%	15.6×
Opus 16 kbps, 5% loss	21.03%	5.23%	4.0×
Opus 8 kbps, 15% loss	26.83%	14.00%	**1.9**×

TABLE VIII
CHUNK-LEVEL EER AND LATENCY VERSUS WINDOW SIZE (CLEAN AUDIO).

Window	AASIST-L EER	Latency
Full utterance	**0.99%**	—
1000 ms	23.62%	55 ms
500 ms	33.28%	55 ms
250 ms	36.43%	58 ms

(Table VIII). AASIST-L requires a fixed 4.04 s input, so shorter chunks are tile-repeated to fill it, discarding most temporal structure—EER rises from 0.99% at utterance level to 36.43% at 250 ms, an accuracy collapse rather than graceful degradation. Adding packet loss on top pushes AASIST-L to 38.70% and the classical baseline past chance (45–55%), which we report as a genuine if unflattering finding. Per-chunk latency remained real-time-capable throughout (≈55 ms, well under any tested window), so the bottleneck is accuracy, not speed: naive chunking of a fixed-input model is not a viable continuous-authentication strategy, and window-aware retraining, not faster hardware, is the needed next step. This independently reproduces the trend reported in [7], where fixed 4 s inputs underperformed full-length inputs (Table II).

G. Limitations

Our real-world evaluation uses eight recordings from a single speaker, device, and room; it demonstrates the failure and a mitigation but does not establish generalization across devices, rooms, or speakers. The attacks are ASVspoof2019 LA (2019-era TTS/VC), so the 54% attack-success figure is likely a lower bound against current cloning tools. VoIP degradation is simulated rather than captured from live calls. Finally, the streaming results use naive tile-repetition rather than a model retrained for short inputs.

V. CONCLUSION AND FUTURE WORK

Speaker verification alone approved over half of the synthetic-voice attacks we tested; a benchmarked anti-spoofing layer closes most of that gap. More consequentially, a model scoring 0.99% EER on its own benchmark failed completely on real-world audio—a gap we diagnosed spectrally and partially closed via targeted domain adaptation at a measured accuracy cost. Future work: scale domain adaptation across more speakers, devices, and rooms; evaluate against modern post-2019 cloning tools; and pursue window-aware retraining for genuinely streaming continuous authentication.

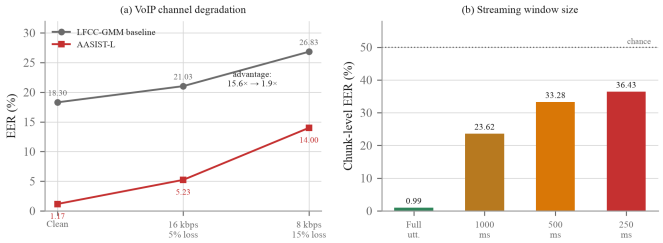


Fig. 4. (a) EER under simulated VoIP channel degradation; AASIST-L's advantage over the classical baseline narrows from 15.6× to 1.9×. (b) Chunk-level EER versus streaming window size on clean audio.

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